

Cryptocurrency Volatility Prediction Using Machine Learning

1. Abstract

This project focuses on predicting cryptocurrency market volatility using machine learning techniques. Historical price data, trading volume, and market capitalization are analyzed to build a predictive model that assists in risk management and decision-making.

2. Problem Statement

Cryptocurrency markets are highly volatile. Predicting volatility helps traders and financial institutions anticipate unstable periods and manage investment risks effectively.

3. Methodology

The project follows a structured machine learning workflow:

- Data preprocessing and cleaning
 - Exploratory data analysis
 - Feature engineering
 - Model training using Random Forest Regressor
 - Performance evaluation using RMSE, MAE, and R^2
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4. Model Implementation

A Random Forest regression model was selected due to its ability to handle non-linear relationships and complex financial data. Feature scaling and proper train-test splitting were applied to improve performance.

5. Results

The trained model achieved reliable prediction accuracy with low error values. Evaluation metrics demonstrate the model's effectiveness in capturing volatility patterns.

6. Conclusion

The project successfully demonstrates the application of machine learning in predicting cryptocurrency volatility. The model can support traders and analysts in proactive risk management.

7. Future Scope

- Use of deep learning models such as LSTM
- Real-time data integration
- Web-based deployment using Streamlit or Flask

8. Final Remarks

This project highlights the importance of data-driven approaches in financial markets and provides a strong foundation for advanced volatility forecasting systems.